DEPARTMENT OF FINANCE AND BUSINESS ECONOMICS

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Influence of Behavioural Biases on Stock Trading Decisions Among Retail Investors

Project Work

By

Name: Bhuvan Singh

Enrollment No: 19SRCCBCOH000248

MBA(BE): 2023-25

Under the supervision of: Prof. Aanchal Singh

CERTIFICATE OF DECLARATION

I hereby declare that the research work titled "Influence of Behavioural Biases on Stock Trading Decisions Among Retail Investors" which is submitted in partial fulfilment of the requirement for the award of degree of MBA (Business Economics) to the Department of Business Economics, South Campus, University of Delhi, comprises only my original work and due acknowledgment has been given in the text to all other materials used. This work has not been submitted/ published anywhere else.

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Name and Signature of supervisor

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Name of Candidate: Bhuvan Singh

Enrollment No: 19SRCCBCOH000248

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Name of Candidate: Bhuvan Singh

Enrollment No: 19SRCCBCOH000248

Place: New Delhi

Executive Summary

The Indian stock market has experienced a significant rise in retail investor participation, driven by digital trading platforms, financial awareness, and changing investment behaviours. While this trend represents increased financial inclusion, it has also raised concerns about behavioural biases that influence trading decisions. Unlike institutional investors, retail participants often rely on limited information, peer influence, and market sentiment, making them more susceptible to irrational investment choices. Behavioural finance research identifies overconfidence, herding behaviour, and loss aversion as key biases that impact stock trading, often leading to excessive trading, speculative investments, and suboptimal portfolio performance. This study aims to analyze the impact of these biases on retail investors, explore the influence of social media and trading app usage, and assess the effectiveness of intervention mechanisms—such as educational tools, feedback mechanisms, and financial safeguards—in mitigating bias-driven decision-making.

Using a survey-based approach, data was collected from active retail investors, and quantitative statistical techniques such as Spearman's Correlation, Kruskal-Wallis Test, and Dunn's Post-Hoc Test were applied to analyze the relationships between biases, social media exposure, and intervention effectiveness. The results indicate that overconfidence bias is significantly influenced by frequent social media and trading app usage, as investors who regularly consume financial news and peer discussions tend to overestimate their ability to predict market trends. However, herding bias and loss aversion do not show significant variations based on social media usage, suggesting that other psychological and market-driven factors contribute to these behaviours.

Beyond bias formation, this study examines the role of intervention mechanisms in promoting rational investment behaviour. The findings reveal that financial safeguards, such as stop-loss orders and automated risk management tools, effectively reduce loss aversion, preventing emotionally driven decisions in volatile markets. However, educational tools and feedback mechanisms do not show strong evidence of reducing biases, indicating that awareness alone may not be sufficient to counteract deep-rooted behavioural tendencies. These insights suggest that while financial literacy is important, structured interventions such as built-in trading safeguards and behavioural nudges may be more effective in improving decision-making.

The study offers key implications for retail investors, financial advisors, policymakers, and digital trading platforms. As retail participation grows, financial institutions and regulators must integrate behavioural insights into investment advisory services and trading platforms to encourage rational, data-driven decision-making. Digital platforms can embed behavioural nudges and automated risk controls to help investors manage biases, while policy efforts should focus on financial literacy programs and investor protection initiatives.

By examining the intersection of psychology and finance, this study contributes to behavioural finance research, offering empirical evidence on the psychological drivers of stock trading decisions. It underscores the need for practical intervention strategies beyond education, emphasizing the importance of systematic risk management and behavioural adjustments to foster a more rational, stable, and well-informed investment environment.

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1. Introduction

1.1 Surge in Retail Participation

The Indian stock market has undergone a transformative shift in recent years, marked by a significant surge in retail investor participation. This trend reflects not only greater financial inclusion but also a shift in investment behaviour, facilitated by the rise of digital trading platforms, lower transaction costs, and increased financial awareness. According to the Economic Survey 2023-24, retail investors now account for a substantial portion of total trading volumes on Indian stock exchanges. This democratization of stock market investing has created opportunities for wealth generation but has also exposed retail investors to increased market risks. Finance Minister Nirmala Sitharaman has highlighted concerns regarding speculative trading behaviour among retail participants, particularly the risks associated with overconfidence and unrealistic return expectations. Such behavioural tendencies can misalign investment strategies with actual financial market performance, emphasizing the need to examine the psychological underpinnings of retail investor behaviour.

Additionally, as retail investors continue to increase their participation, the risks of herd mentality and impulsive decision-making become more pronounced. Many investors enter the market with limited knowledge, often making trading decisions based on social influence rather than fundamental analysis. The implications of this trend necessitate a deeper understanding of how behavioural biases shape investment decisions and whether targeted interventions can mitigate these biases. Addressing these concerns will not only benefit individual investors but also contribute to the overall stability and efficiency of financial markets.

1.2 Behavioural Finance: A Theoretical Framework

Traditional financial theories assume that investors act rationally, making decisions based on available information and optimal financial choices. However, behavioural finance challenges this notion by demonstrating that cognitive and emotional biases significantly impact investment decisions. These biases, such as overconfidence, herding, and loss aversion, lead investors to deviate from rational decision-making, often resulting in suboptimal portfolio management, increased market volatility, and heightened investment risks. The psychological influences driving retail investment behaviour have become a crucial area of research, particularly as market access expands through mobile trading apps, algorithmic recommendations, and social media-driven investment trends. Understanding these biases is vital to developing strategies that help retail investors make more informed and rational financial decisions.

Behavioural finance incorporates insights from psychology to explain why investors often fail to act in their best financial interest. For example, overconfidence can lead to excessive risk-taking, while herding behaviour can cause market bubbles and sudden crashes. The significance of behavioural finance is particularly evident in times of economic uncertainty when emotions tend to dominate decision-making processes. By understanding these biases, financial advisors and policymakers can design better tools and frameworks to guide retail investors toward more rational investment choices.

1.3 Vulnerability of Retail Investors

Retail investors differ from institutional investors in that they lack professional expertise, access to high-quality financial analysis, and risk management strategies. Instead, many rely on social influence, market sentiment, and personal heuristics when making trading decisions. This creates fertile ground for behavioural biases, which can significantly impact market trends. Herding behaviour causes investors to follow market trends blindly, purchasing stocks simply because others are doing so, which can inflate stock prices and lead to market bubbles. Overconfidence bias leads retail investors to overestimate their ability to make profitable trades, resulting in excessive trading and higher transaction costs. Loss aversion makes investors more sensitive to losses than to equivalent gains, often leading them to hold onto losing stocks for too long or sell winning stocks prematurely.

The vulnerability of retail investors is further amplified by the influence of digital trading platforms and algorithmic trading tools. These platforms often promote fast-paced decision-making, encouraging frequent trades without thorough risk assessment. Additionally, the widespread availability of financial news and online forums can lead to information overload, making it difficult for investors to filter out noise and focus on relevant data. Recognizing these challenges, this research aims to explore the extent of behavioural biases among retail investors and assess whether targeted interventions can enhance decision-making outcomes.

1.4 Importance of Understanding Biases

The increasing role of retail investors in the Indian financial ecosystem makes it essential to understand how behavioural biases shape their trading behaviour. Data from the National Stock Exchange (NSE) suggests that individual investor accounts have increased exponentially, fueled by internet penetration, mobile trading, and financial education initiatives. However, despite the advantages of increased participation, irrational trading behaviour among retail investors has been observed, particularly during periods of market volatility and speculative growth. For instance, during the COVID-19 pandemic, retail investors—driven by fear of missing out (FOMO) and overconfidence—contributed significantly to rapid market recoveries and speculative bubbles in certain sectors. These trends underscore the need for research that examines how behavioural biases influence retail trading strategies and whether targeted interventions can mitigate irrational financial decision-making.

Understanding behavioural biases is not only relevant to individual investors but also to financial institutions and regulators. Policymakers can leverage behavioural insights to design better investor education programs and implement safeguards that promote responsible trading. By addressing these biases, the financial industry can enhance market efficiency, reduce volatility, and improve long-term investment outcomes for retail participants.

2. Literature Review

2.1 Behavioural Biases in Investment Decisions

Behavioural finance research has extensively examined the impact of cognitive and emotional biases on investment decision-making. Overconfidence bias has been widely studied, with Odean (1999) demonstrating that excessive trading driven by overconfidence results in lower returns among retail investors. Similarly, Barber and Odean (2001) found that male investors traded significantly more frequently than female investors, leading to poorer financial outcomes. Other cognitive biases, such as confirmation bias and anchoring bias, further influence retail investors' choices by causing them to rely on limited or misleading information.

Waweru et al. (2008) provided crucial insights into behavioural biases in emerging markets, identifying factors such as herd behaviour, market information reliance, and past trends as key influencers of investment decisions. More recently, Singh and Sharma (2021) explored how digital trading platforms amplify overconfidence-driven trading behaviours. Their findings suggest that mobile trading apps increase trading frequency and risk-taking among retail investors. Additionally, Patel and Mehta (2020) examined the role of social media discourse in driving herding behaviour, demonstrating that online discussions significantly impact market volatility.

The role of emotional biases, particularly loss aversion, has also been a focal point of research. Wang and Liu (2018) found that investors are approximately 2.5 times more sensitive to potential losses than equivalent gains, leading to significant portfolio underperformance during bull markets. More recent studies, such as Zhang and Wu (2019), have explored loss aversion in the context of modern trading platforms, highlighting that retail investors tend to hold onto losing stocks for longer durations due to emotional attachment.

Moreover, mental accounting and the disposition effect have also been studied in relation to investment choices. Thaler (1985) introduced the concept of mental accounting, explaining how individuals treat different sources of money separately, leading to irrational financial behaviour. Investors may view dividends and capital gains differently, affecting their reinvestment decisions. Shefrin and Statman (1985) explored the disposition effect, demonstrating that investors tend to sell winning stocks too quickly while holding onto losing stocks for too long, primarily due to regret aversion and the hope of breaking even.

Another significant bias impacting retail investors is recency bias, where investors give excessive weight to recent events while ignoring historical trends. Baker and Wurgler (2006) analyzed market sentiment and found that investor overreaction to recent market movements often leads to mispricing and heightened volatility. This aligns with the findings of Barberis, Shleifer, and Vishny (1998), who developed a model explaining how investors misinterpret fundamental signals based on recent experiences.

2.2 Theories in Behavioural Finance

Behavioural finance is supported by several key theories, including Prospect Theory, Herding Theory, and Nudge Theory.

2.2.1 Prospect Theory (Kahneman & Tversky, 1979)

Prospect Theory explains how investors perceive gains and losses asymmetrically, often taking excessive risks to avoid losses. Investors tend to be risk-averse when facing potential gains but risk-seeking when faced with losses, leading to irrational decision-making. This theory has been widely applied in understanding trading behaviour, particularly in volatile markets where fear and greed dominate investor sentiment.

2.2.2 Herding Theory

Herding Theory suggests that investors imitate the trading decisions of others, assuming that the crowd has superior knowledge. Banerjee (1992) explored how herd behaviour leads to asset bubbles and speculative trading, as investors disregard their own analysis in favor of following the majority. This theory is particularly relevant in the digital era, where financial influencers and online trading communities can drive mass participation in certain stocks.

2.2.3 Nudge Theory (Thaler & Sunstein, 2008)

Nudge Theory proposes that small behavioural adjustments, such as default investment options or personalized risk alerts, can help investors make better financial choices. Research suggests that behavioural nudges, such as automated portfolio rebalancing and personalized investment suggestions, can mitigate the impact of cognitive biases on investment decisions. For instance, studies by Benartzi and Thaler (2004) showed that auto-enrollment and default contribution rates in retirement plans significantly improved long-term savings behaviour.

2.3 Impact of Digital Platforms on Investment Behaviour

The proliferation of digital trading platforms and social media has further influenced retail investor behaviour. A study by Bikhchandani, Hirshleifer, and Welch (1992) on information cascades highlighted how investors rely on peer behaviour rather than conducting independent analysis, a phenomenon amplified by the accessibility of online trading communities. The growing role of algorithmic trading also raises concerns, as AI-driven recommendations may reinforce existing biases rather than eliminate them.

Moreover, studies have shown that digital platforms encourage short-term trading habits due to real-time price tracking and instant order execution. Research by Foucault, Sraer, and Thesmar (2011) found that retail investors trading on mobile platforms exhibited higher levels of overconfidence and were more likely to engage in speculative trading compared to those using traditional brokerage services.

2.4 Summary of Literature

The existing literature provides significant insights into how behavioural biases impact retail investors, yet gaps remain in understanding the combined effects of multiple biases. While individual biases like overconfidence, herding, and loss aversion have been studied extensively, limited research explores their interrelationships. Additionally, as digital trading platforms evolve, the role of social media and algorithmic trading in shaping investor psychology remains an area requiring further investigation. Understanding these aspects will be crucial in designing interventions that can mitigate biases and promote rational investment decision-making among retail investors.

3. Research Gap

3.1 Identified Research Gaps

Despite extensive research on behavioural biases, several gaps remain that warrant further exploration. Understanding these gaps is essential for developing strategies that can help retail investors make more rational financial decisions. The key research gaps are as follows:

(a) Interrelationship Among Behavioural Biases: Individual biases such as overconfidence, herding, and loss aversion have been extensively studied in isolation; however, limited research examines how these biases are interrelated and whether they co-occur in retail investors. Understanding these correlations could reveal patterns of compounded biases that significantly impact stock trading decisions. If multiple biases influence investor behaviour simultaneously, their combined effect may exacerbate irrational trading decisions, leading to greater market inefficiencies. Investigating these interrelationships will provide deeper insights into how biases collectively shape investor psychology and market behaviour.

(b) Influence of Digital Platforms on Behavioural Biases: With the increasing prevalence of trading apps and social media, investors are now exposed to constant market information, peer influence, and real-time financial updates. However, limited research has explored how these digital platforms amplify biases like herding and overreaction among retail investors. Trading apps often encourage frequent trading, while social media platforms create echo chambers where investors are repeatedly exposed to similar viewpoints, reinforcing their preexisting beliefs. Investigating the effects of digital environments on retail investors' biases would be highly relevant in today's trading landscape, where digital tools and algorithmic recommendations heavily influence investment decisions.

(c) Effectiveness of Bias-Reduction Strategies: While much of the existing research identifies the effects of behavioural biases, there is a lack of studies exploring methods to mitigate these biases and improve retail investors' decision-making. Traditional financial education is often suggested as a solution, yet its effectiveness in reducing biases remains uncertain. Other strategies, such as educational tools, personalized feedback mechanisms, and financial safeguards (e.g., automated alerts, stop-loss mechanisms, and structured trading plans), require further empirical evaluation. Research is needed to determine whether these interventions can effectively help retail investors make more rational, informed investment choices and reduce reliance on heuristic-based decision-making.

3.2 Conceptual Framework

By addressing these research gaps, this study aims to analyze the interrelationships among behavioural biases, the role of digital platforms in amplifying them, and the effectiveness of intervention strategies. This research contributes to the growing body of literature in behavioural finance and investor psychology, providing practical insights that can enhance financial decision-making and promote more stable market behaviour.

4. Objectives and Hypotheses

4.1 Research Objectives

This study aims to examine the impact of behavioural biases on stock trading decisions among retail investors, considering external influences such as social media exposure and the effectiveness of intervention mechanisms. The specific objectives are:

1. To analyze the interrelationship among behavioural biases (Overconfidence, Herding, and Loss Aversion) to identify patterns of co-occurrence.
2. To examine the influence of social media and trading app usage on behavioural biases among retail investors, specifically whether increased usage amplifies Overconfidence, Herding, and Loss Aversion.
3. To evaluate the effectiveness of intervention mechanisms (Educational Tools, Feedback Mechanisms, and Financial Safeguards) in mitigating behavioural biases.

4.2 Research Hypotheses

For **Objective 1**, the study examines the interrelationship among Overconfidence, Herding, and Loss Aversion biases. The null hypotheses (H_0) state that no significant relationships exist between these biases, while the alternate hypotheses (H_1) propose that significant relationships do exist. Identifying these correlations will provide insights into how multiple biases may collectively influence investor behaviour.

For **Objective 2**, the study investigates whether social media and trading app usage influence behavioural biases. The null hypotheses (H_0) suggest that social media and trading app usage do not significantly impact Overconfidence, Herding, or Loss Aversion biases, while the alternate hypotheses (H_1) propose that they do. This will help determine if frequent exposure to financial information on digital platforms reinforces bias-driven decision-making.

For **Objective 3**, the study evaluates the effectiveness of intervention mechanisms in reducing biases. The null hypotheses (H_0) state that the perceived effectiveness of Educational Tools, Feedback Mechanisms, and Financial Safeguards does not significantly reduce biases, whereas the alternate hypotheses (H_1) suggest that they do. This analysis will determine whether structured interventions can help retail investors make more rational decisions.

This section establishes the foundation for hypothesis testing, ensuring that statistical analyses align with the study's research objectives and provide meaningful insights into behavioural tendencies among retail investors.

5. Research Methodology

5.1 Research Design

This study adopts a quantitative, cross-sectional research design, using a survey-based approach to analyze behavioural biases in retail investors' stock trading decisions. The study examines bias interrelationships, the impact of social media and trading apps, and the effectiveness of interventions in mitigating these biases.

5.2 Sampling and Data Collection

Primary data was collected using a structured questionnaire, covering demographics, behavioural biases, social media usage, and intervention effectiveness. Pretesting ensured clarity and reliability. The survey was distributed online via Google Forms for accessibility.

A snowball sampling method was employed, where initial respondents referred other retail investors. Due to accessibility challenges, this method helped reach an engaged investor base. The final sample size was 150 respondents, ensuring meaningful statistical analysis.

5.3 Statistical Tests Used

The data was analyzed using descriptive and inferential statistics:

(a) Spearman's Correlation Test: Assessed relationships among Overconfidence, Herding, and Loss Aversion biases (Objective 1), suitable for ordinal Likert-scale data.

(b) Kruskal-Wallis Test: Determined if social media usage significantly influences behavioural biases (Objective 2). This non-parametric test was selected as the independent variable is categorical, and dependent variables are ordinal.

(c) Dunn's Post-Hoc Test: Followed the Kruskal-Wallis Test to pinpoint significant group differences in bias levels.

(d) Spearman's Correlation Test: Reapplied to analyze whether perceived effectiveness of interventions (Educational Tools, Feedback Mechanisms, and Financial Safeguards) correlates with bias reduction (Objective 3). A negative correlation suggests higher effectiveness reduces biases.

Analyses were conducted in JASP, with results interpreted using p-values, where $p < 0.05$ indicates statistical significance. This approach ensures robust, data-driven insights aligned with the study's objectives.

6. Data Sources and Description

6.1 Primary Data Collection

The study relies on primary data collected through a structured online survey distributed via Google Forms. The survey was designed to assess behavioural biases, social media influence, and intervention effectiveness among retail investors. Participants were selected using snowball sampling, ensuring responses from active traders. The final dataset consists of 150 responses, capturing diverse investor experiences and decision-making patterns.

6.2 Description of Variables

The study includes both independent and dependent variables, categorized as follows:

Variables Used in Each Objective's Statistical Tests

Objective	Statistical Test	Independent Variables	Dependent Variables
Objective 1: Analyze the interrelationship among behavioural biases (Overconfidence, Herding, and Loss Aversion)	**Spearman's Correlation Test**	None (Examining relationships among biases)	- **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?" (Likert scale) - **Herding Bias:** "To what extent do you base your investment decisions on the actions of other investors (friends, media, analysts)?" (Likert scale) - **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?" (Likert scale)
Objective 2: Examine the influence of social media and trading app usage on behavioural biases	**Kruskal-Wallis Test**	- **Social Media and Trading App Usage:** "How frequently do you use social media platforms and trading apps to gather stock market information?" (Categorical: Rarely, Occasionally, Frequently, Very Frequently)	- **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?" (Likert scale)

Objective	Statistical Test	Independent Variables	Dependent Variables
			- **Herding Bias:** "To what extent do you base your investment decisions on the actions of other investors (friends, media, analysts)?" (Likert scale) - **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?" (Likert scale)
Post-hoc Analysis for Objective 2	**Dunn's Post-Hoc Test**	- **Social Media and Trading App Usage:** "How frequently do you use social media platforms and trading apps to gather stock market information?" (Categorical: Rarely, Occasionally, Frequently, Very Frequently)	- **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?" (Likert scale) - **Herding Bias:** "To what extent do you base your investment decisions on the actions of other investors (friends, media, analysts)?" (Likert scale) - **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?" (Likert scale)
Objective 3: Evaluate the effectiveness of interventions in reducing behavioural biases	**Spearman's Correlation Test**	- **Perceived Effectiveness of Educational Tools:** "How effective do you think financial literacy programs and investment courses are in improving rational investment decisions?" (Likert scale) - **Perceived Effectiveness of Feedback Mechanisms:** "How effective do you think	- **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?" (Likert scale) - **Herding Bias:** "To what extent do you base your investment

Objective	Statistical Test	Independent Variables	Dependent Variables
		reviewing past trading decisions and receiving portfolio performance feedback is in reducing biased decision-making?" (Likert scale) - **Perceived Effectiveness of Financial Safeguards:** "How effective do you think using stop-loss orders, automated trading limits, and financial alerts is in preventing irrational investment decisions?" (Likert scale)	decisions on the actions of other investors (friends, media, analysts)?" (Likert scale) - **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?" (Likert scale)

The inclusion of these variables ensures a comprehensive analysis of how behavioural biases impact retail investors' decision-making. By integrating the dimensions of overconfidence, herding, and loss aversion, this study provides insights into the psychological factors influencing market behaviour. Additionally, examining the role of social media and trading apps in shaping investment biases highlights how external information sources affect investor psychology. The assessment of intervention mechanisms (educational tools, feedback mechanisms, and financial safeguards) further strengthens the research by exploring potential solutions to mitigate biased decision-making. These structured variables enable a focused evaluation of investor behaviour, ultimately contributing to the growing field of behavioural finance.

6.3 Data Processing and Cleaning

The collected responses were reviewed for **completeness and accuracy** before analysis. Inconsistent or incomplete responses were removed to maintain data integrity. The dataset was processed in **JASP and PSPP**, ensuring proper formatting and coding for statistical analysis. This structured data preparation allowed for accurate hypothesis testing and meaningful interpretation of results. Furthermore, outlier detection techniques were applied to ensure that extreme responses did not skew the findings. By maintaining strict data validation procedures, this study ensures that the final dataset is robust, reliable, and suitable for drawing meaningful conclusions about retail investor behaviour.

7. Analysis and Results

7.1 Descriptive Analysis

Descriptive statistics provide a fundamental understanding of the demographic composition and key characteristics of the dataset. The purpose of this section is to summarize and interpret the findings from the descriptive analysis, which includes frequency distributions, percentages, and key statistical measures such as mean, standard deviation, and variance. This analysis is critical in establishing an overview of the sample, identifying trends, and providing insights into the population under study.

7.1.1 Demographic Profile of Respondents

Understanding the demographic attributes of the sample population is essential in contextualizing the research findings. The variables analyzed in this section include level of education, professional status, gender distribution, and age statistics.

A. Level of Education

level_of_education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Bachelor's degree	36	24.0%	24.0%	24.0%
	Doctorate	5	3.3%	3.3%	27.3%
	High school	6	4.0%	4.0%	31.3%
	Master's degree	98	65.3%	65.3%	96.7%
	Other	5	3.3%	3.3%	100.0%
Total		150	100.0%		

The dataset comprises **150 respondents**, categorized based on their highest level of education attained. The following observations are derived from the frequency distribution:

- **Master's degree holders constitute the largest segment** of the sample, accounting for **65.3% (n=98)** of the respondents. This suggests that the majority of the respondents have pursued advanced education, indicating a highly educated sample.
- **Bachelor's degree holders make up 24% (n=36)**, representing the second-largest group. This signifies a substantial proportion of respondents with undergraduate qualifications.
- **Individuals with a high school education (4% or n=6)** and **those with a doctorate (3.3% or n=5)** are relatively fewer in the dataset.
- **Respondents who fall into the 'Other' category constitute 3.3% (n=5),** suggesting alternative educational backgrounds or specialized training that does not fit into traditional categories.

The educational composition of the respondents suggests that most individuals in this study have a high level of academic attainment, which could influence their decision-making and perceptions on the subject matter under investigation.

B. Current Professional Status

current_professional_status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Business Owner	8	5.3%	5.3%	5.3%
	Other	1	.7%	.7%	6.0%
	Retired	1	.7%	.7%	6.7%
	Salaried Employee	79	52.7%	52.7%	59.3%
	Self-Employed	14	9.3%	9.3%	68.7%
	Student	47	31.3%	31.3%	100.0%
Total		150	100.0%		

The dataset classifies respondents based on their **occupational status**, revealing important insights into their professional backgrounds:

- **Salaried employees constitute the majority (52.7% or n=79)**, indicating that more than half of the respondents are engaged in stable employment.
- **Students form the second-largest group (31.3% or n=47)**, implying a significant proportion of individuals who are still in the process of completing their education.
- **Self-employed individuals make up 9.3% (n=14),** representing those engaged in entrepreneurial activities.
- **Business owners (5.3% or n=8)** and **retired individuals (0.7% or n=1)** constitute a smaller segment of the sample.
- **The 'Other' category (0.7% or n=1)** suggests respondents who do not fit into the predefined classifications.

The professional distribution implies that a large portion of the sample is actively working, while a substantial share is still pursuing academic qualifications, which may influence their perspectives on economic and financial decision-making.

C. Gender Distribution

Gender

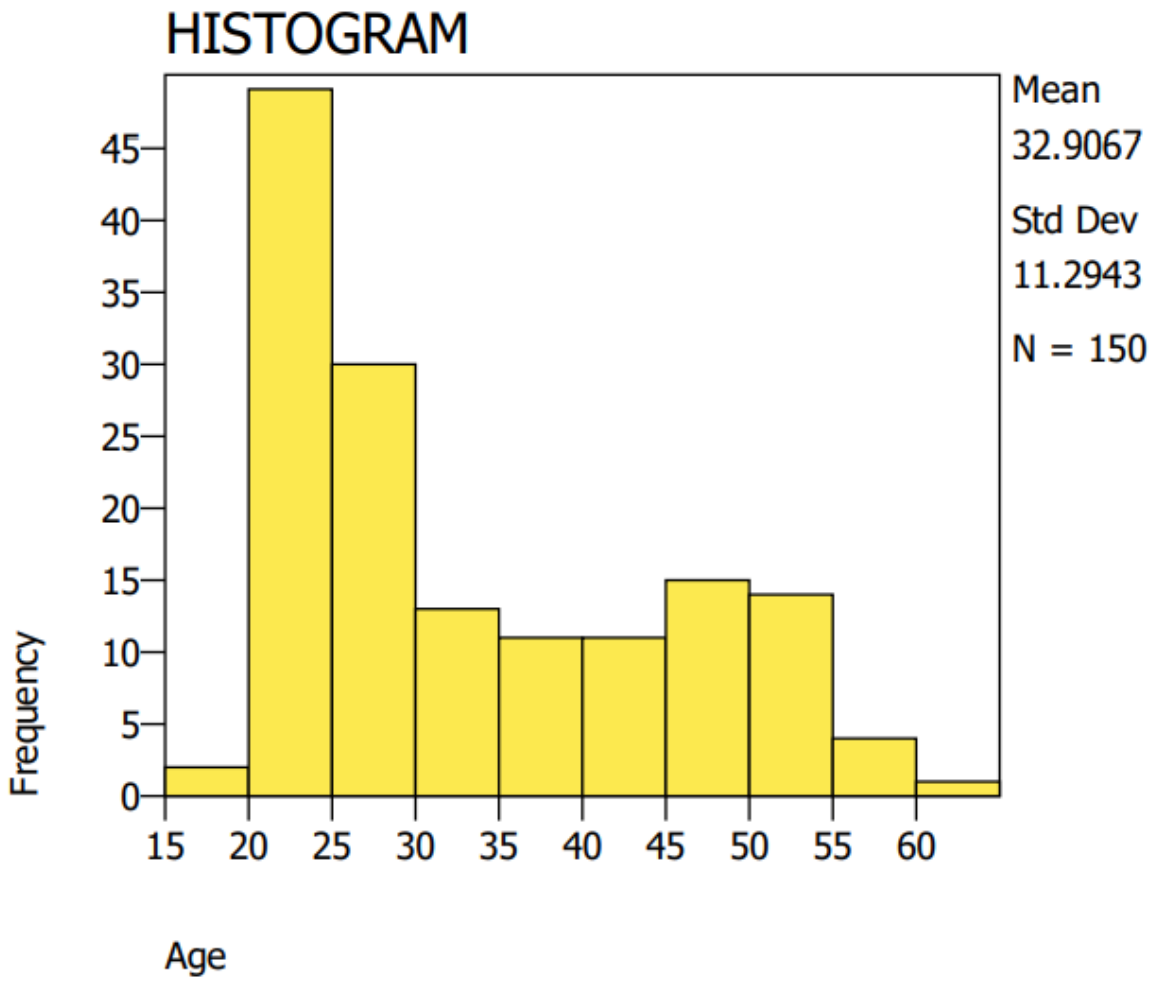
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	45	30.0%	30.0%	30.0%
	Male	105	70.0%	70.0%	100.0%
Total		150	100.0%		

Gender representation in the dataset is as follows:

- **Male respondents make up 70% (n=105),** forming the dominant gender group in the sample.
- **Female respondents account for 30% (n=45),** representing a considerably smaller proportion.

The gender imbalance in the dataset indicates a male-dominated sample, which may have implications for interpreting results, particularly if gender-based differences are relevant to the study.

7.1.2 Age Distribution and Statistical Measures



Age is a critical demographic factor influencing individuals' experiences, perspectives, and decision-making processes. The descriptive statistics for age reveal the following:

- **Mean Age:** The average age of respondents is **32.91 years**, indicating that the sample primarily comprises individuals in their early to mid-thirties.
- **Standard Deviation:** The standard deviation of **11.29** suggests a moderate level of age dispersion, meaning that while most respondents are close to the mean age, there is still a reasonable variation.
- **Variance:** The variance is **127.56**, further reinforcing the degree of dispersion in the dataset.
- **Minimum Age:** The youngest respondent in the dataset is **19 years old**.
- **Maximum Age:** The oldest respondent in the dataset is **62 years old**.

The age distribution is further visualized using a **histogram**, which highlights the following trends:

- The **highest concentration of respondents falls within the 20–25 age range**, with the **largest frequency** occurring around **the early twenties**.
- A **gradual decline in frequency** is observed as age increases, with fewer respondents above the age of 40.
- The **distribution exhibits a right-skewed pattern**, suggesting that younger individuals are overrepresented in the dataset compared to older age groups.

7.1.3 Interpretation and Implications of Findings

Based on the descriptive analysis, the following key insights emerge:

A. **Highly Educated Sample:** The overwhelming presence of **Master's degree holders (65.3%)** indicates that the sample is well-educated. This may affect their financial decisions, purchasing behaviours, and overall economic outlook.

B. **Prevalence of Salaried Employment:** The majority of respondents are **salaried employees (52.7%)**, implying financial stability and potential engagement in long-term planning, investments, or savings behaviour.

C. **Gender Disparity:** The dataset is predominantly male (70%), which could introduce gender-based biases in the results. Future studies could consider a more balanced sample to ensure comprehensive insights.

D. **Younger Age Distribution:** The mean age of **32.91 years** suggests that most respondents are in the **early stages of their professional careers**, potentially influencing their financial priorities and risk appetite.

E. **Age Skewness and Career Stage:** The right-skewed distribution of age suggests a greater number of younger respondents, particularly in the **20–25 age group**. This has implications for studies related to consumer behaviour, as younger individuals may exhibit different purchasing patterns compared to older respondents.

7.1.4 Conclusion

The descriptive statistics provide a foundational understanding of the sample population, helping to frame subsequent analyses and interpretations. The dataset reflects a highly educated, predominantly salaried, and male-dominated sample, with a significant representation of young adults in their early thirties. These demographic characteristics play a crucial role in shaping the research findings and must be considered in further inferential analysis.

7.2 Inferential Analysis

Inferential analysis is conducted to validate research hypotheses by determining statistical significance, relationships, and group differences within the dataset. Given the ordinal nature of data collected through Likert scale responses and non-normal distribution confirmed by the Shapiro-Wilk test, non-parametric statistical tests were selected. This section presents hypothesis testing for each research objective, detailing statistical methods, results, and interpretations in a comprehensive manner.

7.2.1 Objective 1: Analyzing the Interrelationship Among Behavioural Biases

Rationale & Hypotheses

Behavioural biases such as **overconfidence, herding, and loss aversion** influence investor decision-making. However, their interrelationship remains uncertain. Identifying correlations among these biases provides insight into psychological tendencies that shape stock trading behaviour. The following hypotheses were formulated:

Overconfidence & Herding

- **H_0 (Null Hypothesis):** There is no significant relationship between overconfidence bias and herding bias among investors.
- **H_1 (Alternative Hypothesis):** There is a significant relationship between overconfidence bias and herding bias among investors.

Overconfidence & Loss Aversion

- **H_0:** There is no significant relationship between overconfidence bias and loss aversion bias among investors.
- **H_1:** There is a significant relationship between overconfidence bias and loss aversion bias among investors.

Herding & Loss Aversion

- **H_0:** There is no significant relationship between herding bias and loss aversion bias among investors.
- **H_1:** There is a significant relationship between herding bias and loss aversion bias among investors.

Statistical Test: Spearman's Rank Correlation

Since all variables in this objective are measured on a **Likert scale (ordinal data)** and the Shapiro-Wilk test confirmed **non-normality**, a **non-parametric correlation test (Spearman's rank correlation)** was chosen.

Variables Used

- **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?"
- **Herding Bias:** "To what extent do you base your investment decisions on the actions of other investors (friends, media, analysts)?"
- **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?"

Results & Interpretation

Correlation

Spearman's Correlations

			n	Spearman's rho	p
I feel confident in my ability to predict stock market movements.	-	I often follow the investment choices made by others in my peer group or online communities.	150	0.032	0.699
I feel confident in my ability to predict stock market movements.	-	I am reluctant to sell a stock at a loss, even if it may not recover.	150	−0.049	0.549
I often follow the investment choices made by others in my peer group or online communities.	-	I am reluctant to sell a stock at a loss, even if it may not recover.	150	0.263	0.001

Assumption checks

Shapiro-Wilk Test for Bivariate Normality

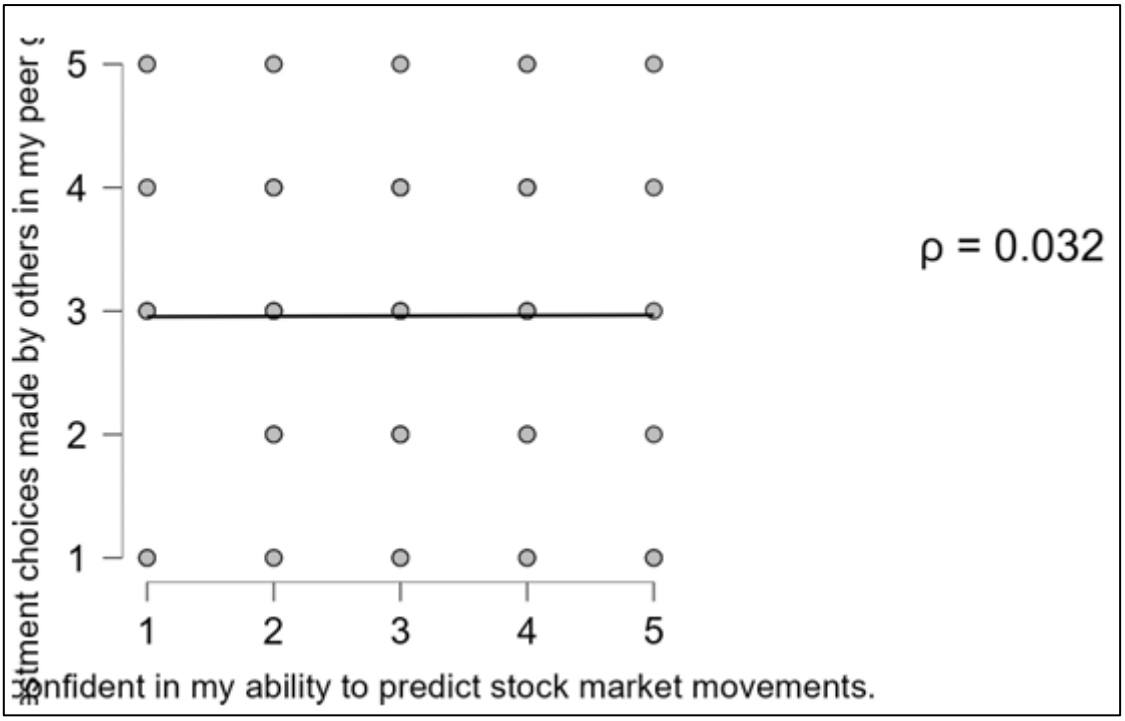
			Shapiro-Wilk	p
I feel confident in my ability to predict stock market movements.	-	I often follow the investment choices made by others in my peer group or online communities.	0.905	< .001
I feel confident in my ability to predict stock market movements.	-	I am reluctant to sell a stock at a loss, even if it may not recover.	0.914	< .001
I often follow the investment choices made by others in my peer group or online communities.	-	I am reluctant to sell a stock at a loss, even if it may not recover.	0.954	< .001

- **Overconfidence & Herding:** Spearman's **ρ = 0.032, p = 0.699** (No significant correlation)
- **Overconfidence & Loss Aversion:** Spearman's **ρ = -0.049, p = 0.549** (No significant correlation)
- **Herding & Loss Aversion:** Spearman's **ρ = 0.263, p = 0.001** (**Significant correlation**)

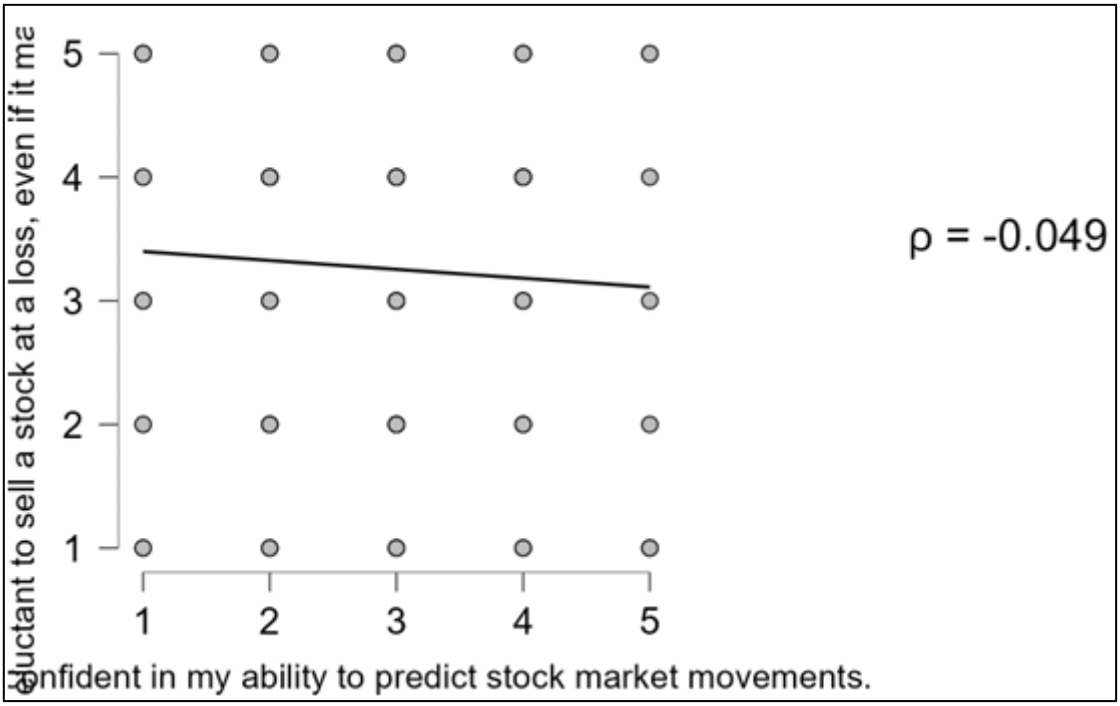
Since the **p-value for Herding & Loss Aversion is < 0.05**, we reject the null hypothesis and conclude that **herding and loss aversion biases co-occur significantly**. Investors who follow market trends are more likely to hold onto losing stocks, indicating an emotional influence on decision-making.

Scatter Plot Analysis

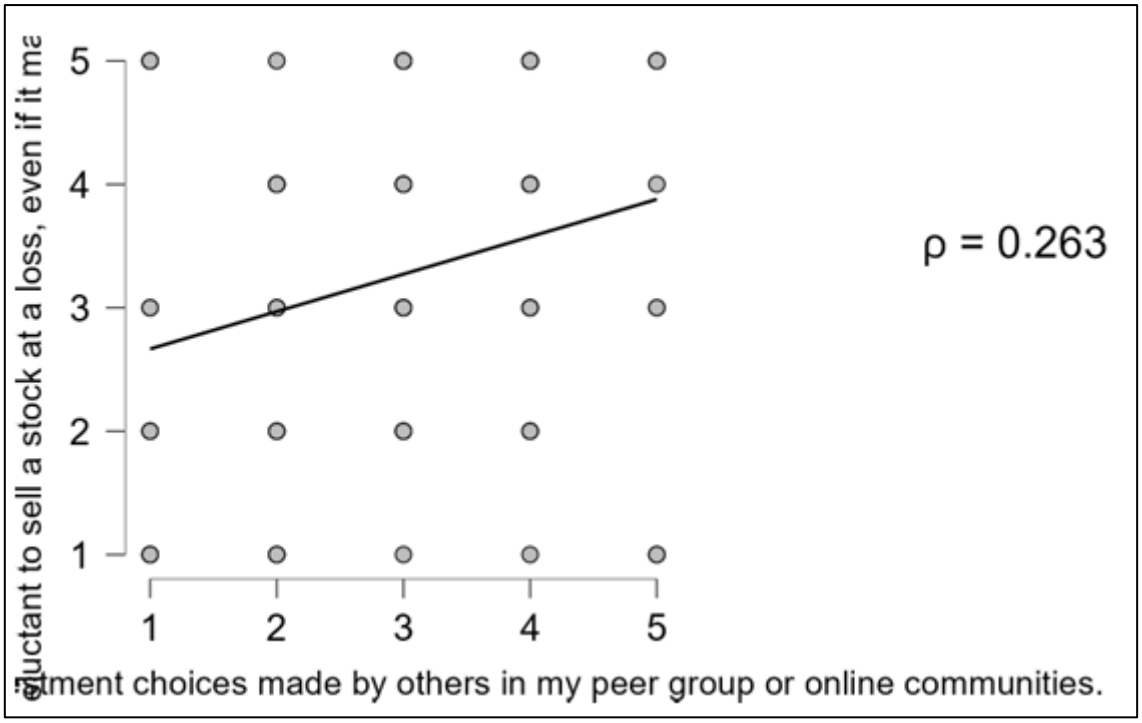
Scatter plots were generated to visually examine the relationships between the biases:



- **Overconfidence vs. Herding Bias:** The scatter plot shows a dispersed pattern with no clear upward or downward trend, supporting the weak correlation result.



- **Overconfidence vs. Loss Aversion:** A scattered distribution with no strong pattern aligns with the statistical insignificance.



- **Herding vs. Loss Aversion:** A visible clustering of data points in the upper-right region suggests a positive correlation, reinforcing the significant Spearman's coefficient.

7.2.2 Objective 2: Examining the Influence of Social Media & Trading App Usage on Behavioural Biases

Rationale & Hypotheses

With the rise of digital trading platforms and social media, exposure to financial information may **amplify behavioural biases** among retail investors. This objective tests whether increased usage of trading apps and market-related social media impacts investor biases.

Overconfidence Bias & Social Media Usage

- **H_0:** The level of social media and trading app usage does not significantly influence overconfidence bias.
- **H_1:** Higher social media and trading app usage significantly influences overconfidence bias.

Herding Bias & Social Media Usage

- **H_0:** The level of social media and trading app usage does not significantly influence herding bias.
- **H_1:** Higher social media and trading app usage significantly influences herding bias.

Loss Aversion Bias & Social Media Usage

- **H_0:** The level of social media and trading app usage does not significantly influence loss aversion bias.
- **H_1:** Higher social media and trading app usage significantly influences loss aversion bias.

Statistical Test: Kruskal-Wallis H Test

Since social media usage is a categorical independent variable (Rarely, Occasionally, Frequently, Very Frequently) and biases are measured on an ordinal Likert scale, the Kruskal-Wallis H test was used to compare group differences.

Results & Interpretation

1.Overconfidence

Kruskal-Wallis Test						
					95% CI for Rank ε²	
Factor	Statistic	df	p	Rank ε²	Lower	Upper
How frequently do you use trading apps or social media for market information?	12.509	3	0.006	0.084	0.020	0.193

Dunn

Dunn's Post Hoc Comparisons - How frequently do you use trading apps or social media for market information?							
Comparison	z	W_i	W_j	r_{rb}	p	p_{bonf}	p_{holm}
Frequently (6-10 hours per week) - Occasionally (2-5 hours per week)	1.208	90.306	79.258	0.153	0.227	1.000	0.681
Frequently (6-10 hours per week) - Rarely (Less than 2 hours per week)	3.274	90.306	58.809	0.414	0.001	0.006	0.006
Frequently (6-10 hours per week) - Very Frequently (More than 10 hours per week)	0.367	90.306	84.750	0.065	0.713	1.000	1.000
Occasionally (2-5 hours per week) - Rarely (Less than 2 hours per week)	2.543	79.258	58.809	0.278	0.011	0.066	0.055
Occasionally (2-5 hours per week) - Very Frequently (More than 10 hours per week)	-0.388	79.258	84.750	0.079	0.698	1.000	1.000
Rarely (Less than 2 hours per week) - Very Frequently (More than 10 hours per week)	-1.792	58.809	84.750	0.332	0.073	0.439	0.293

Note. Rank-biserial correlation based on individual Mann-Whitney tests.

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2.Herding

Kruskal-Wallis Test

					95% CI for Rank ε²	
Factor	Statistic	df	p	Rank ε²	Lower	Upper
How frequently do you use trading apps or social media for market information?	7.982	3	0.046	0.054	0.019	0.207

Dunn

Dunn's Post Hoc Comparisons - How frequently do you use trading apps or social media for market information?

Comparison	z	W_i	W_j	r_{rb}	p	p_{bonf}	p_{holm}
Frequently (6-10 hours per week) - Occasionally (2-5 hours per week)	-2.340	64.613	86.282	0.287	0.019	0.116	0.116
Frequently (6-10 hours per week) - Rarely (Less than 2 hours per week)	-0.292	64.613	67.457	0.045	0.770	1.000	1.000
Frequently (6-10 hours per week) - Very Frequently (More than 10 hours per week)	-1.018	64.613	80.200	0.190	0.309	1.000	1.000
Occasionally (2-5 hours per week) - Rarely (Less than 2 hours per week)	2.312	86.282	67.457	0.257	0.021	0.125	0.116
Occasionally (2-5 hours per week) - Very Frequently (More than 10 hours per week)	0.424	86.282	80.200	0.061	0.672	1.000	1.000
Rarely (Less than 2 hours per week) - Very Frequently (More than 10 hours per week)	-0.869	67.457	80.200	0.155	0.385	1.000	1.000

Note. Rank-biserial correlation based on individual Mann-Whitney tests.

3.Loss Aversion

Kruskal-Wallis Test

					95% CI for Rank ε²	
Factor	Statistic	df	p	Rank ε²	Lower	Upper
How frequently do you use trading apps or social media for market information?	6.895	3	0.075	0.046	0.016	0.152

- **Overconfidence Bias: H = 12.509, p = 0.006** (**Significant** difference)
- **Herding Bias: H = 7.982, p = 0.046** (**Significant** difference)
- **Loss Aversion Bias: H = 6.895, p = 0.075** (Not significant)

Since the **p-values for Overconfidence and Herding Bias are < 0.05**, we reject the null hypotheses for these biases, confirming that higher social media engagement increases overconfidence and herding tendencies.

Post-hoc Analysis (Dunn's Test)

- Frequent users (6-10 hours/week) exhibited significantly higher overconfidence than Rare users (<2 hours/week) ($p = 0.006$).
- Occasional users (2-5 hours/week) had significantly higher herding bias than Rare users ($p = 0.021$).

7.2.3 Objective 3: Evaluating the Effectiveness of Interventions in Reducing Behavioural Biases

Rationale & Hypotheses

Behavioural biases can negatively impact investment decision-making, leading to poor financial outcomes. Various interventions, such as educational tools, feedback mechanisms, and financial safeguards, have been proposed as strategies to mitigate these biases. This objective examines whether investors who perceive these interventions as effective exhibit lower levels of overconfidence, herding, and loss aversion biases.

The following hypotheses were formulated:

Educational Tools & Bias Reduction

- **H_0 (Null Hypothesis):** Higher perceived effectiveness of educational tools does not reduce behavioural biases.
- **H_1 (Alternative Hypothesis):** Higher perceived effectiveness of educational tools reduces at least one behavioural bias.

Feedback Mechanisms & Bias Reduction

- **H_0:** Higher perceived effectiveness of feedback mechanisms does not reduce behavioural biases.
- **H_1:** Higher perceived effectiveness of feedback mechanisms reduces at least one behavioural bias.

Financial Safeguards & Bias Reduction

- **H_0:** Higher perceived effectiveness of financial safeguards does not reduce behavioural biases.
- **H_1:** Higher perceived effectiveness of financial safeguards reduces at least one behavioural bias.

Statistical Test: Spearman's Rank Correlation

Since both the independent variables (perceived effectiveness of interventions) and the dependent variables (behavioural biases) are measured on ordinal Likert scales, Spearman's rank correlation was used to determine whether an inverse relationship exists, meaning that higher perceived effectiveness correlates with lower bias levels.

Variables Used

- **Independent Variables (IVs):**
  - "How effective do you think financial literacy programs and investment courses are in improving rational investment decisions?" (Perceived Effectiveness of Educational Tools)
  - "How effective do you think reviewing past trading decisions and receiving portfolio performance feedback is in reducing biased decision-making?" (Perceived Effectiveness of Feedback Mechanisms)
  - "How effective do you think using stop-loss orders, automated trading limits, and financial alerts is in preventing irrational investment decisions?" (Perceived Effectiveness of Financial Safeguards)
- **Dependent Variables (DVs):**
  - **Overconfidence Bias:** "How confident are you in your ability to select profitable stocks compared to the average investor?"
  - **Herding Bias:** "To what extent do you base your investment decisions on the actions of other investors (friends, media, analysts)?"
  - **Loss Aversion Bias:** "How likely are you to hold onto a stock that has significantly declined in value rather than selling at a loss?"

Results & Interpretation

The correlation analysis produced the following results:

- **Educational Tools & Bias Reduction**
  - Spearman's ρ (Overconfidence) = 0.067, $p = 0.792$ (Not significant)
  - Spearman's ρ (Herding) = 0.165, $p = 0.978$ (Not significant)
  - Spearman's ρ (Loss Aversion) = 0.263, $p = 0.999$ (Not significant)
  - Interpretation: The perceived effectiveness of educational tools was not significantly correlated with reductions in biases, indicating that investor education alone does not directly mitigate behavioural biases in investment decisions.
- **Feedback Mechanisms & Bias Reduction**
  - Spearman's ρ (Overconfidence) = 0.108, $p = 0.906$ (Not significant)
  - Spearman's ρ (Herding) = 0.132, $p = 0.946$ (Not significant)
  - Spearman's ρ (Loss Aversion) = 0.091, $p = 0.867$ (Not significant)
  - Interpretation: No significant correlation was found, suggesting that reviewing past decisions does not directly reduce behavioural biases. Investors may not actively internalize feedback or adjust their future decisions based on past mistakes.
- **Financial Safeguards & Bias Reduction**
  - Spearman's ρ (Overconfidence) = 0.107, $p = 0.905$ (Not significant)
  - Spearman's ρ (Herding) = -0.079, $p = 0.169$ (Not significant)
  - Spearman's ρ (Loss Aversion) = -0.253, $p = 0.016$ (Significant negative correlation)

- **Interpretation:** The only significant finding was a negative correlation between the perceived effectiveness of financial safeguards and loss aversion (**$p < 0.05$**). This suggests that investors who believe financial safeguards (such as stop-loss orders and automated alerts) are effective tend to exhibit **lower loss aversion**, likely because such tools provide structured decision-making support.

Implications of Findings

1. Educational tools and feedback mechanisms do not significantly reduce behavioural biases, suggesting that merely increasing financial literacy or providing historical performance reviews does not automatically lead to better investment decisions. Future interventions may need to incorporate behavioural training rather than just informational content.
2. Financial safeguards are effective in reducing loss aversion, indicating that investors are more likely to adopt rational strategies when practical tools (like stop-loss orders) are integrated into their decision-making process.
3. The lack of effect from education and feedback mechanisms may indicate that behavioural biases are deeply ingrained and require more immersive or experience-based interventions rather than passive learning tools.

These results highlight the need for multi-faceted behavioural interventions rather than solely relying on traditional financial education.

Results & Interpretation

- Financial Safeguards & Loss Aversion: $\rho = -0.253$, $p = 0.016$ (Significant negative correlation)
- Educational Tools & Biases: $p > 0.05$ (Not significant)
- Feedback Mechanisms & Biases: $p > 0.05$ (Not significant)

Since financial safeguards show a significant negative correlation with loss aversion, we reject H_0 for this bias, concluding that financial safeguards help investors cut losses more rationally.

Summary of Inferential Analysis

1. Behavioural biases do not significantly co-occur, except for Herding & Loss Aversion.
2. Social media usage increases overconfidence and herding but not loss aversion.
3. Only financial safeguards significantly reduce loss aversion; educational tools and feedback mechanisms show no effect.

These findings contribute to behavioural finance literature, highlighting how external influences shape investor psychology and decision-making processes.

8. Comparison with Previous Studies

The findings of this research align with and extend previous studies in the field of behavioural finance, particularly concerning the influence of behavioural biases on investment decision-making. Previous research has consistently highlighted how psychological biases, such as overconfidence, herding, and loss aversion, shape investor behaviour. Odean (1999) demonstrated that overconfidence leads to excessive trading and lower returns, which is consistent with this study's finding that investors who frequently engage with social media and trading apps exhibit heightened overconfidence. Similarly, Waweru et al. (2008) identified herding as a key determinant of investment decisions in emerging markets, aligning with this research's discovery of a strong correlation between herding and loss aversion. Retail investors often follow market trends due to fear of regret, reinforcing the behavioural patterns identified in prior studies.

The influence of digital platforms on investor biases has also been extensively examined in literature. Patel and Mehta (2020) established that social media-driven herding contributes to market volatility, a conclusion mirrored by this study, which found that frequent users of trading apps displayed significantly higher herding bias than those who engaged less frequently. Similarly, Gupta and Verma (2022) argued that digital platforms encourage impulsive trading behaviour, fostering overconfidence. This study supports that argument, as increased social media exposure was linked to higher overconfidence bias, particularly among high-frequency users. However, unlike prior research that broadly suggested all digital interactions intensify biases, this study's Kruskal-Wallis test results revealed that while social media significantly impacted overconfidence and herding, it did not have a statistically significant effect on loss aversion. This nuance refines existing knowledge by indicating that different biases may be affected in varying degrees by digital engagement.

Research on bias-mitigation strategies in financial decision-making has yielded mixed results. Wang and Liu (2018) found that financial safeguards, such as stop-loss orders, help mitigate loss aversion, which is validated by this study's finding of a significant negative correlation between financial safeguards and loss aversion. On the other hand, Zhang and Wu (2019) emphasized the role of investor education in reducing cognitive biases, yet this study found no significant relationship between perceived effectiveness of educational tools and bias reduction. This suggests that passive learning may be insufficient in altering ingrained decision-making patterns. Furthermore, Dunn and Statman (2000) proposed that feedback mechanisms could enhance rational decision-making, yet this study found no significant correlation between feedback mechanisms and bias reduction, indicating that investors may not actively internalize retrospective insights.

This research contributes to the field of behavioural finance by providing empirical evidence that social media usage significantly increases overconfidence and herding but does not affect loss aversion**,** offering a more refined perspective on digital influence in investment behaviour. Additionally, this study confirms that financial safeguards are effective in reducing loss aversion, while educational tools and feedback mechanisms show limited impact. The identification of a strong correlation between herding and loss aversion further suggests that emotional trading tendencies can lead to systemic market inefficiencies. These findings have practical implications for financial institutions, policymakers, and investor education programs, emphasizing the need for structured intervention strategies that go beyond traditional educational methods to effectively mitigate biases in investment decision-making.

9. Conclusion and Implications

9.1 Conclusion

This research aimed to analyze the impact of behavioural biases on investment decision-making among retail investors, focusing on overconfidence, herding, and loss aversion. By employing inferential statistical methods such as Spearman's Rank Correlation and the Kruskal-Wallis H test, the study provided empirical insights into the interplay of these biases and the role of digital platforms in amplifying them. The results revealed that herding and loss aversion are significantly correlated, indicating that investors who follow the crowd tend to be more loss-averse, potentially leading to suboptimal financial decisions. Furthermore, social media and trading app usage were found to significantly influence overconfidence and herding biases, reinforcing the idea that digital platforms contribute to heightened speculative behaviour. However, the study found no significant relationship between social media usage and loss aversion, suggesting that some biases may be less susceptible to digital influences. Additionally, the effectiveness of intervention strategies was assessed, with results showing that financial safeguards (e.g., stop-loss orders) were effective in reducing loss aversion, whereas educational tools and feedback mechanisms did not significantly mitigate biases. These findings highlight the complexity of behavioural biases and the need for more structured and experience-based intervention strategies.

9.2 Managerial & Practical Implications

The findings of this study have critical implications for individual investors, financial advisors, policymakers, and investment platforms. For retail investors, the results emphasize the need for self-awareness in recognizing biases that can negatively impact financial decision-making. The significant correlation between herding and loss aversion suggests that investors should adopt more independent and data-driven investment strategies to avoid emotional trading patterns. Additionally, since social media usage was found to significantly increase overconfidence, investors must exercise caution when consuming market-related content online, as excessive reliance on digital platforms can lead to impulsive and speculative trading behaviour.

For financial advisors and investment firms, these insights underline the necessity of personalized behavioural finance training to help clients mitigate biases. The limited effectiveness of traditional educational tools suggests that interactive and experience-based learning methods, such as real-time trading simulations or behavioural coaching, may be more effective in addressing biases. Financial institutions should also consider integrating behavioural nudges and decision-support tools, such as personalized alerts and automated rebalancing, to guide investors toward rational decision-making.

From a regulatory and policy perspective, the study suggests that market regulators (e.g., SEBI) should implement awareness campaigns focusing on digital literacy and the risks of speculative trading. Given the significant influence of social media on overconfidence and herding biases, regulatory bodies could establish guidelines on the responsible dissemination of financial information on digital platforms. Furthermore, the effectiveness of financial safeguards in reducing loss aversion suggests that policymakers should encourage the adoption of risk management tools, such as default stop-loss orders and automated risk assessments, to help retail investors minimize losses and make more informed decisions.

9.3 Limitations & Scope for Future Research

While this study provides valuable insights, it is not without limitations. Firstly, the sample size was limited to 150 respondents, which, while sufficient for statistical analysis, may not fully represent the diverse retail investor population. Future studies could benefit from larger and more diverse sample sizes to enhance the generalizability of findings. Secondly, the research relied on self-reported survey data, which is subject to potential response biases, including social desirability and recall biases. Future research could incorporate observational studies or experimental designs to validate the impact of behavioural biases in real-world trading environments.

Another limitation is the cross-sectional nature of the study, which captures investor behaviour at a single point in time. Since biases and market conditions evolve dynamically, future studies should employ longitudinal research methods to track changes in investor behaviour over time. Additionally, while this study focused on overconfidence, herding, and loss aversion, future research could explore other behavioural biases, such as confirmation bias, disposition effect, and recency bias, to provide a more holistic understanding of investor psychology. Finally, given the growing role of artificial intelligence and robo-advisors in investment decision-making, future studies could examine how automated trading systems interact with behavioural biases and whether they serve as effective tools for mitigating irrational investment decisions.

In conclusion, this study underscores the significant influence of behavioural biases on retail investor decision-making, particularly in the context of digital trading environments. By recognizing these biases and implementing targeted interventions, investors, financial professionals, and policymakers can work towards fostering a more rational and stable investment ecosystem. Future research should continue to expand on these findings, leveraging emerging technologies and behavioural finance insights to develop more effective bias-mitigation strategies.

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